

Riley Beenders

AI PRODUCT ENGINEERING | USER WORKFLOWS | AUTOMATION

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PROFESSIONAL SUMMARY

Product-focused R&D and engineering leader with 10+ years of multidisciplinary experience turning user needs into practical systems. Owns products end to end across product judgment, design, automation, user-facing operations, commercialization, technical support, and continuous improvement in complex real-world environments.

CORE EXPERTISE

Product Engineering: Technical product development; problem definition; user workflows and interfaces; product judgment; commercialization; stakeholder feedback; lifecycle ownership

AI & Automation: AI-assisted diagnostics; machine-vision integration; workflow automation; LabVIEW/Modbus; Python; local LLM tooling (LM Studio); Git/GitHub

Enterprise Delivery: ERP/CRM workflows; Kubernetes; Docker and containerized applications; monitoring and diagnostics; risk analysis; technical documentation and training

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA

January 2025 - Present

Lead, Research & Development Engineer

- Led ICARUS-Lite from problem definition through mechanical, electrical, and software design, commercialization, technical support, and after-sales service.
- Shaped and implemented the PROTEOR Print user experience with an external software partner, combining remote operation, live video, AI-assisted diagnostics, and production metrics.
- Built an automated LabVIEW/Modbus QC workflow that converts networked sensor data into live tolerance views and traceable PDF reports for customers and field support.
- Managed Microsoft Business Central ERP implementation for the Coplay site, translating local product and operating needs into workflows shared with Proteor's global locations.

Filament Innovations | Whitehall, PA

May 2020 - January 2025

CTO | Lead Electro-Mechanical Engineer | Industrial Engineer | Engineering Technician

- Progressed through four roles to own the product-development pipeline and technology strategy while remaining hands-on across software, firmware, electrical, mechanical, and manufacturing systems.
- Developed and hosted a redundant Kubernetes-based ERP/CRM that centralized orders, inventory, tasks, projects, and customer interactions and connected dynamic BOMs to purchasing.
- Designed and launched Gen3 POSEIDON, combining industrial-servo motion, safety controls, firmware, hybrid extrusion, and a production-oriented customer platform.
- Delivered the custom KRAKEN system from customer requirements through CAD, BOMs, sourcing, fabrication, assembly, firmware tuning, operator training, and ongoing support.

ADDITIONAL RELEVANT EXPERIENCE

RJB.Engineering - Owner & Engineering Consultant | 2015 - Present | Deliver project-based engineering support for automotive clients and operate a small online hardware-accessory store.

Northampton Area School District - Computer/Network Technician | 2016 - 2019 | Built infrastructure and created an Arduino tool that accelerated deployment of 2,500+ user devices.

EDUCATION & PROFESSIONAL DEVELOPMENT

Penn State University | B.S., Electro-Mechanical Engineering

December 2024

Relevant development: Machine Learning, MATLAB, Simulink, and Simscape Onramps (2026); Lean Six Sigma Yellow Belt (2025); Penn State Certificate in Engineering Design and Tools (2023)